AP CALCULUS AB

Unit 1 • Lesson 1.2 notes

Lesson 1.2 • Defining Limits and Using Limit Notation

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Read and write limit notation, including one-sided limits.
- Decide whether a limit exists from its one-sided limits.

Vocabulary. limit, left-hand limit, right-hand limit, does not exist (DNE).

1. What a limit means

Key idea

$\lim_{x\to c} f(x) = L$ means that $f(x)$ gets arbitrarily close to L for all x sufficiently close to c, with $x \neq c$. The value $f(c)$, and even whether it exists, plays no role.

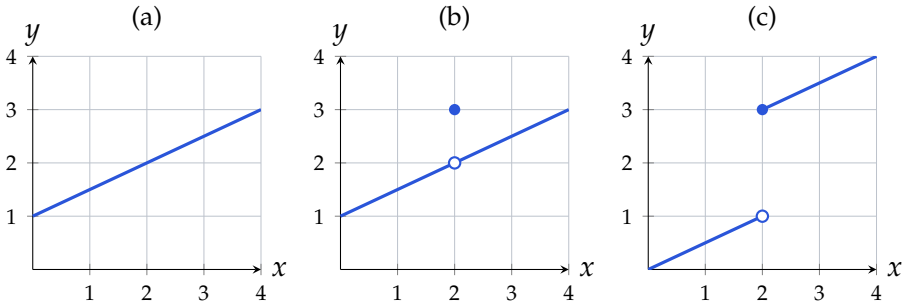
2. One-sided limits

$\lim_{x\to c^-} f(x)$ uses only $x < c$; $\lim_{x\to c^+} f(x)$ uses only $x > c$.

Existence

$\lim_{x\to c} f(x)$ exists and equals L **if and only if** $\lim_{x\to c^-} f(x) = L$ and $\lim_{x\to c^+} f(x) = L$.

Example 1 • Three situations at $x = 2$



Solution. (a) $\lim_{x\to 2} f(x) = 2 = f(2)$. (b) $\lim_{x\to 2} f(x) = 2$ although $f(2) = 3$. (c) Left limit 1, right limit 3: the limit does not exist.

Example 2 • Piecewise formula

$$g(x) = \begin{cases} x^2 - 1, & x < 3 \\ 2x + 2, & x \geq 3. \end{cases}$$

Solution. $\lim_{x\to 3^-} g(x) = 8$ and $\lim_{x\to 3^+} g(x) = 8$, so $\lim_{x\to 3} g(x) = 8$.

Common mistakes

- Answering $\lim_{x\to c} f(x) = f(c)$ automatically. This is true only when f is continuous at c.
- Saying a limit "equals DNE". Write "the limit does not exist".

Quick check

1. $\lim_{x \to 1^-} h(x) = 5$ and $\lim_{x \to 1^+} h(x) = 5$; $h(1) = 0$. Find $\lim_{x \to 1} h(x)$.
2. Can $\lim_{x \to 4} f(x)$ exist if $f(4)$ is undefined?

Answers. 1. 5. 2. Yes; for example $\frac{x^2-16}{x-4} \to 8$.